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Class Analytics



2.20 Leaves and Photosynthesis

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Written by: Douglas Wilkin, Ph.D. | Jean Brainard, Ph.D.

Fact-checked by: The CK-12 Editorial Team

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What you will learn

- Structure and function of different kinds of specialized **tissues** in leaves - **mesophyll**, **veins**, and **epidermis**
- Role of **stomata** and **guard cells** to control **transpiration** and **gas exchange**



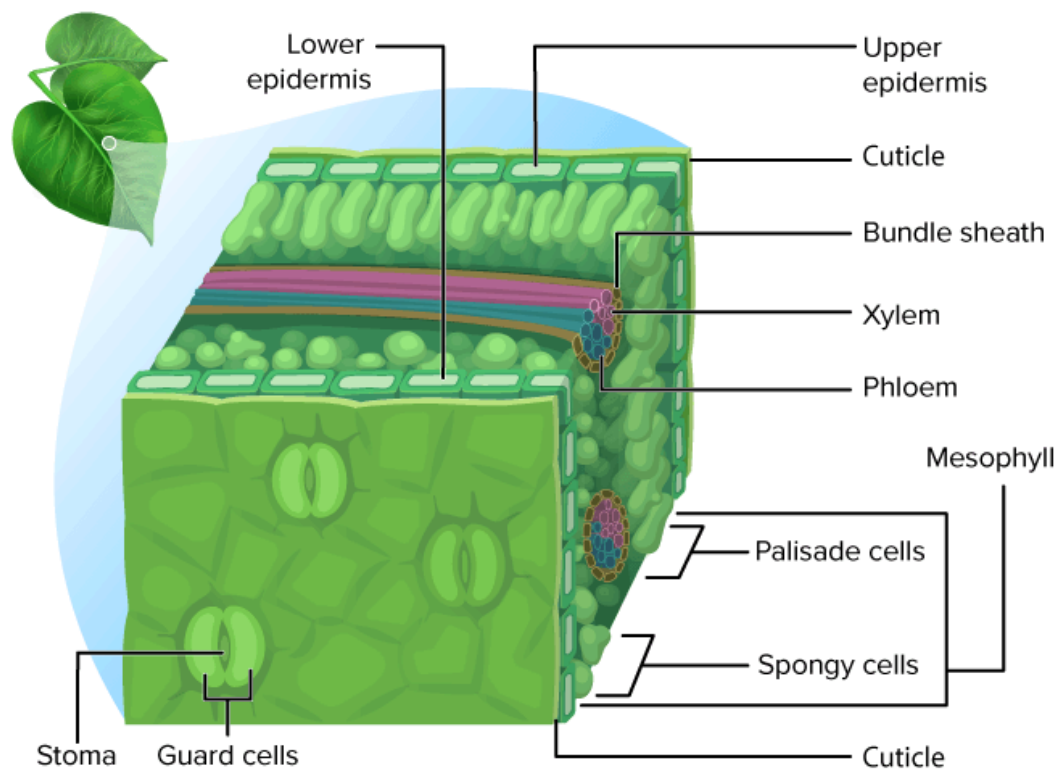
[Figure 1]

Oxygen — the oxygen that we breathe — is just a waste product of what reaction?

Every split second that sunlight hits that leaf, **photosynthesis** is initiated, bringing **energy** into the **cell system**. It could be said that this is one of the most important - if not the absolutely most important - **chemical reactions**. And it all starts with the leaf.

Photosynthesis is the process that uses energy from the sun, together with carbon dioxide and **water**, to make **glucose** and oxygen. The primary role of photosynthesis is to make the **carbohydrate**, suggesting that oxygen, which is released back into the **atmosphere**, is just a waste product.

You can think of a single leaf as a photosynthesis factory. A factory has specialized machines to produce a product. It's also connected to a transportation system that supplies it with raw materials and carries away the finished product. In all these ways, a leaf resembles a factory. The cross section of a leaf in the **Figure below** lets you look inside a leaf "factory."

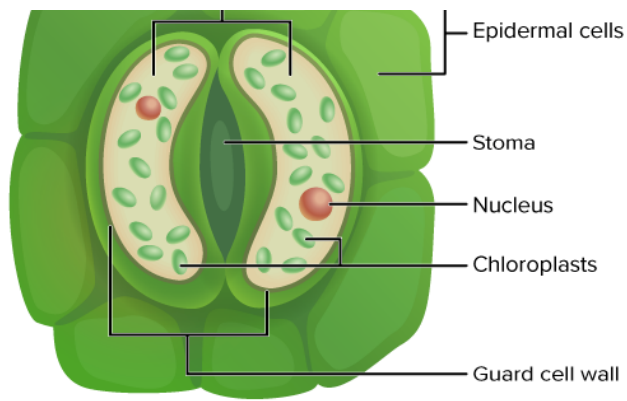


[Figure 2]

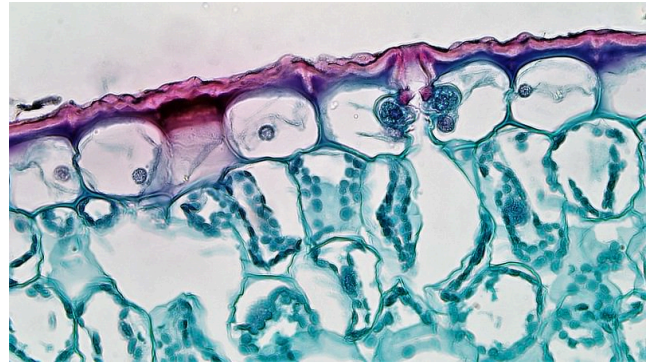
There's more to a leaf than meets the eye. Can you identify the functions of each of the labeled structures in the diagram?

A leaf consists of several different kinds of specialized tissues that work together to make **food** by photosynthesis. The major tissues are mesophyll, veins, and epidermis.

- **Mesophyll** makes up most of the leaf's interior. This is where photosynthesis occurs. Mesophyll consists mainly of parenchyma with **chloroplasts**.
- **Veins** are made primarily of **xylem** and **phloem**. They transport water and **minerals** to the **cells** of leaves and carry away dissolved sugar.
- The **epidermis** of the leaf consists of a single layer of tightly-packed dermal cells. They secrete waxy **cuticle** to prevent **evaporation** of water from the leaf. The epidermis has tiny pores called **stomata** (singular, stoma) that control transpiration and gas exchange with the air (see **Figure below**). For photosynthesis, stomata must control the transpiration of water vapor and the exchange of carbon dioxide and oxygen. Stomata are flanked by **guard cells** that swell or shrink by taking in or losing water through **osmosis**. When they do, they open or close the stomata (see **Figure below**).



[Figure 3]



[Figure 4]

(first) Stomata are flanked by guard cells that swell or shrink by taking in or losing water through osmosis. (second) Stomata and guard cells of a xerophyte leaf.

Structure Of The Leaf | Plant | Biology | The FuseSchool

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DID YOU KNOW?

Not all plant **species** have leaves that are green in color, some have red, yellow, or orange leaves. There are three kinds of primary pigments in the plant leaves: **chlorophyll** (green color), carotenoids (yellow or yellow-orange color), and anthocyanins (red or purple color). The color of the leaf is determined by the pigment which is present in the highest concentration.

Summary

- Specialized cells and tissues in leaves work together to perform photosynthesis.
- The major tissues in a leaf are mesophyll, veins, and epidermis.

Review

1. Explain how a leaf is like a factory.
2. Explain the role of stomata during photosynthesis.
3. What controls the opening of stomata?



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